Introduction

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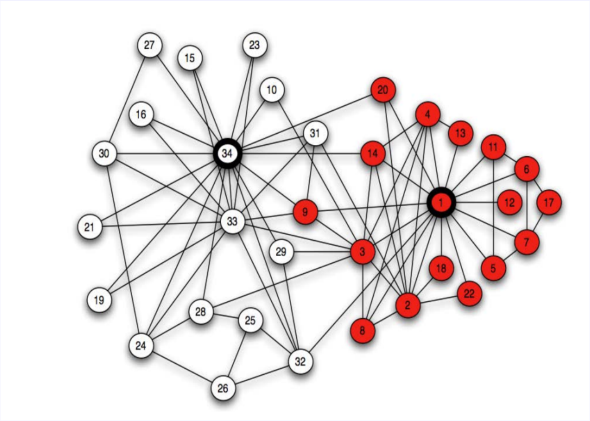
What are Networks

- Network is a collection of units and connections among them.
- This "unit" and "connections" can mean many different things in different contexts. But there are some commonalities as well.
  - Epidemics: individuals, spreading virus
  - Friendship: individuals, personal relationships
  - Acquaintance: individuals, information exchanges
  - Membership: individuals/firms, membership
  - Financial Network: Financial institutions, credit flow
  - Weblink: Websites, direct links
  - Patent citation network: Patents, citations
  - ...

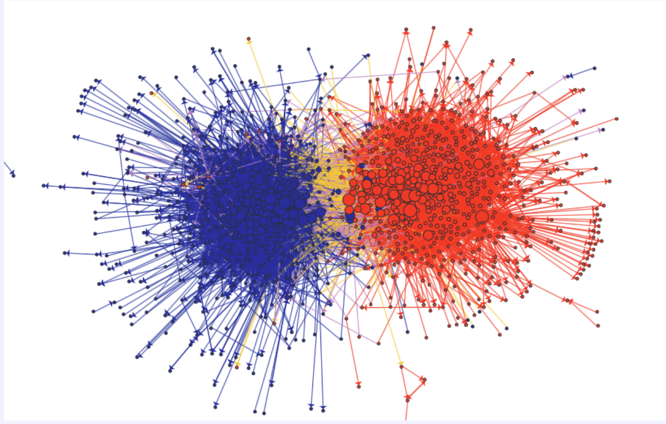
- Our focus is on social and economic networks (but not always). Each node is often an individual/firm and each link represents some interactions among them.
- The important feature of social/economics networks is that those individuals/firms make a decision. So, it is important to understand how they acquire information through network, process information, and make a decision based on them.

Example 1



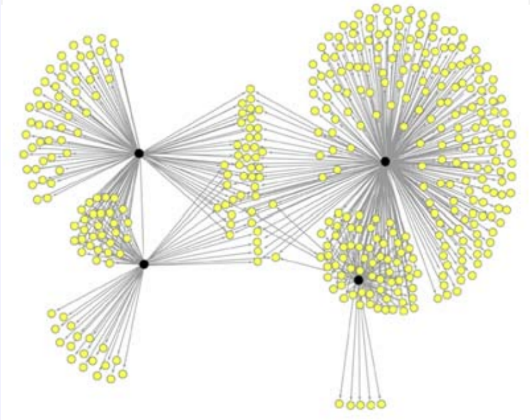
Social network of friendships within 34-person karate club, Zachary (1977)

Example 2



Links between political blogs before 2004 presidential election, Adamic and Glance (2005)

Example 3



E-mail recommendations for Japanese graphic novel, Leskovec, Adamic, and Huberman (2007)

Why Network?

- Networks are everywhere in our society.
- We could not really see them before. But now various network data is available.
- Let's see a few well-known examples, old and new, to see why they may be worth a close look.

Small-World Phenomenon

We say "small world" as we often find unexpected connections to people we didn't know/first met.

There is a word "six degrees of separation," which means that you can reach any random person by six steps in following your friends, and the friends of your friends...

How closely connected are we?

Small World Experiment (Milgram, 1968)

- Randomly selected 296 residents in some cities in Kansas and Nebraska try forwarding a letter to the target contact in Boston.
- 64 letters reached the target and the average steps it took among them was $5.5 - 6$.

Many similar experiments found this magic number.

How should we interpret this?

Small World Phenomenon

Here is another example of “small world”.

- **Erdős Number**
  - Paul Erdős was a very prolific Hungarian mathematician. He published about 1500 papers, many of them co-authored.
  - A researcher’s Erdős number is the distance to Erdős in the co-authorship network.
  - It is often surprisingly small (mine is 4).

When can this happen? For what kind of network? Is that a good thing?

Strong and Weak Ties

- As a part of his Ph.D. thesis, Mark Granovetter, a sociologist, interviewed those who changed their jobs recently and asked where they got information about their new job.
- They tend to learn about a new opportunity from their acquaintances (weak ties) rather than their close friends (strong ties). Hence “The Strength of Weak Ties” (1973).
- Idea: your close friends do not know what you do not know. Your less close acquaintances bring fresh information from the outside of your local community (this was in 60’s). A “bridge” to the outside world tends to be not so close friend.

Interesting perspective on diffusion of information.

Diffusion and Intervention

- Virus spread through contact network.
- How to stop contagion? Where should we intervene with limited resource?
- We may want to target who spreads virus more than others, someone who plays a “central” role in diffusion.
- Maybe people with more connections (doctors)?

Diffusion of Information

Banerjee, Chandrasekhar, Duflo, and Jackson (2012, 2019)

- You like to disseminate information about a new product/project through a word of mouth. Who should you target?
- Some villagers in Indivan villages are asked to spread information about a new program (Microfiance, Immunization).
- They found that some type of "centrality" of seed villagers is a good predictor of villagers' participation in the program.
- In field experiments, villagers are first asked who should be a seed. They actually picked who are "central" in theoretical sense without the knowledge about the entire network structure.

Implication for viral marketing?

A Brief History

- A theory of social network is developed in Sociology through the 20th century (Granovetter, Milgram in 60s and 70s)
- Graph theory dates back to 18th century (Seven Bridges of Königsberg, Euler). Random Graph by Erdős and Rènyi, Gilbert in 50s.
- In Economics, simple model of "network effect" has been used since 70s-80s, where each economic agent is affected by the average behavior. Have started to adopt more structural network theory since 90s.

What We (Try To) Cover

Here is a preliminary list of topics I plan to cover (maybe a change the exact order may be different). The chapter numbers are suggested readings from "Networks, Crowds, and Markets".

- Diffusion: Epidemic, Diffusion of behavior (Ch19, Ch 21)
- Opinion/Belief dynamics
- Bayesian Social Learning (Ch16)
- Networked Markets (Ch10, Ch 11)
- Network Effects (Ch17)
- Network Formation (Ch14)

What We (Try To) Cover

To study, those issues, we need the following as a tool

- Graph Theory and Social Networks (Ch2, Ch3, Ch4)
- Random Graph (Ch20)
- Game Theory (Ch6)